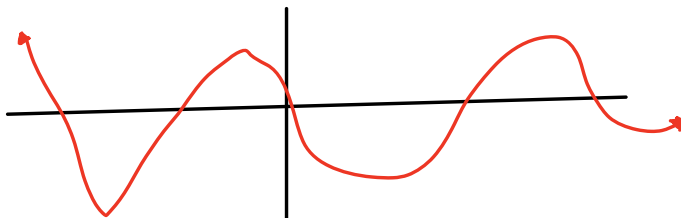
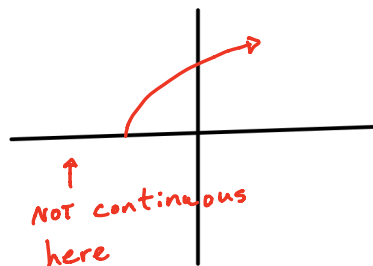
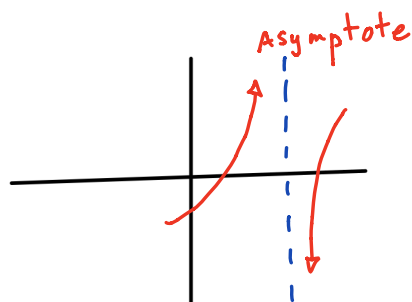
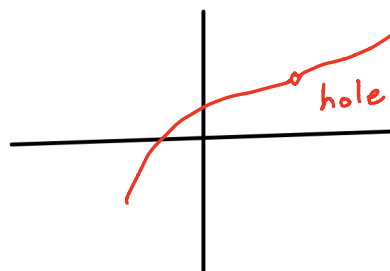
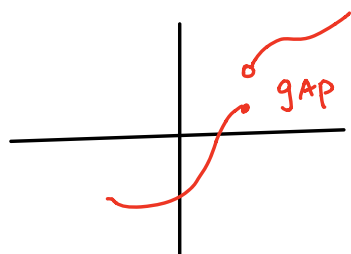


Section 2.5 Continuous Functions



This is a continuous function.

The following functions are NOT continuous.



Def $f(x)$ is continuous at $x=a$ means

$$\lim_{x \rightarrow a} f(x) = f(a)$$

↑
This means that the left & right sides agree.

↑
This makes sure there is no hole at $x=a$.

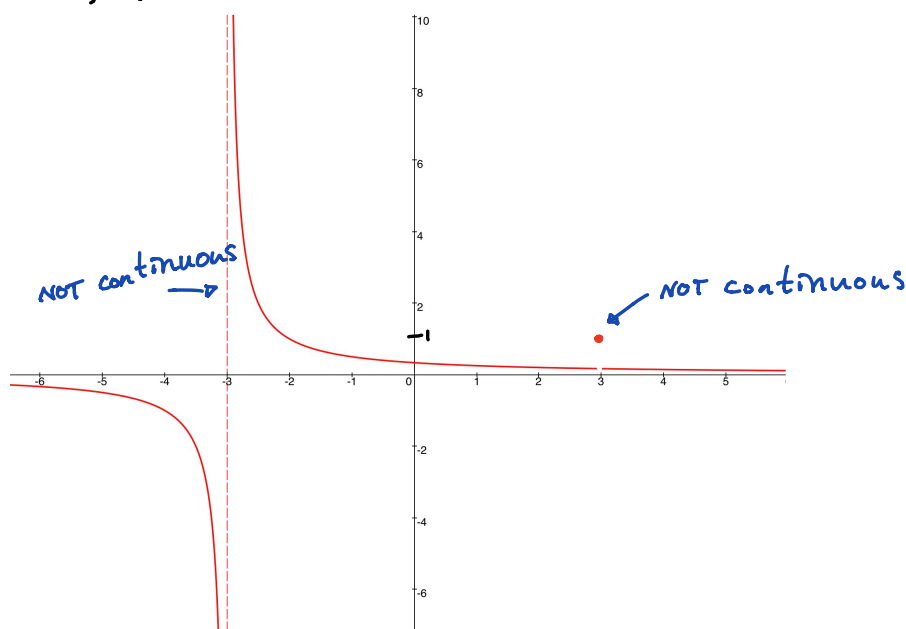
Ex ① Is $f(x)$ continuous?

$$f(x) = \begin{cases} \frac{x-3}{x^2-9} & \text{if } x \neq \pm 3 \\ 1 & \text{if } x = \pm 3 \end{cases}$$

$$\lim_{x \rightarrow 3} \frac{x-3}{x^2-9} = \lim_{x \rightarrow 3} \frac{x-3}{(x-3)(x+3)} = \lim_{x \rightarrow 3} \frac{1}{x+3} = \frac{1}{6}$$

Since $\frac{1}{6}$ is NOT the same as 1,

$f(x)$ is NOT continuous at $x=3$.

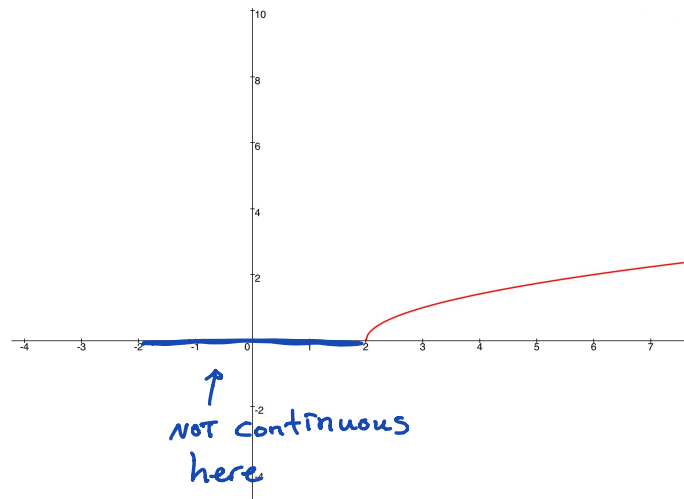


Ex ② Is $g(x)$ continuous over $[-2, 2]$?

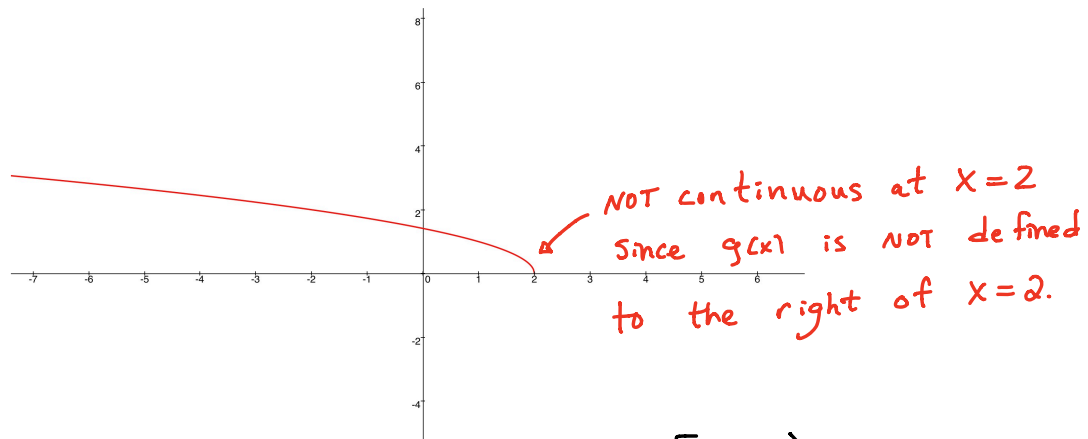
$$A) \quad g(x) = \sqrt{x-2}$$

$$g(-2) = \sqrt{-2-2} = \sqrt{-4} \quad \text{which is NOT a real number.}$$

So, $g(x)$ is NOT continuous on $[-2, 2]$



B) $g(x) = \sqrt{2-x}$

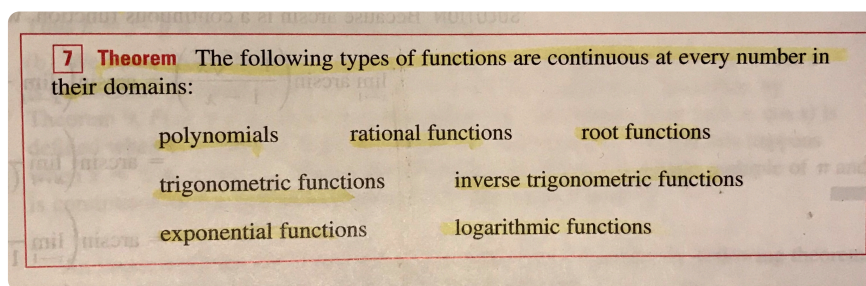


$g(x)$ is continuous on $[-2, 2)$,
but NOT continuous on $[-2, 2]$.

Ex ③ Where is $f(x)$ continuous?

$$f(x) = \sqrt[5]{x} \cdot (1+x^2)$$

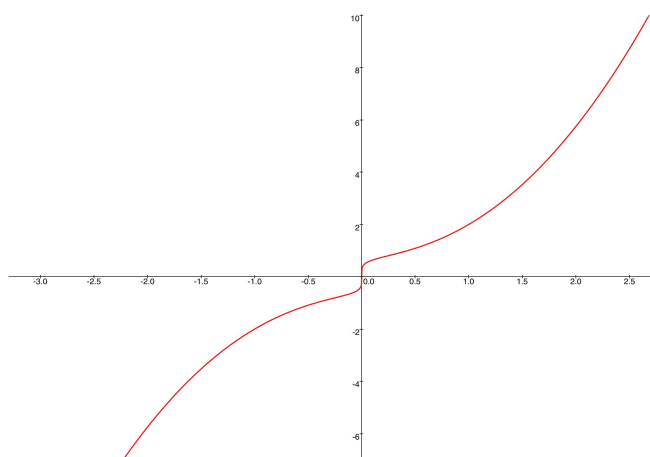
See pg. 123



$1+x^2$ is a polynomial, so it is continuous.

$\sqrt[5]{x}$ has domain $(-\infty, \infty)$, so it is continuous.

By Theorem 7, $f(x) = \sqrt[5]{x}(1+x^2)$ is continuous.

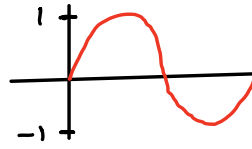


This function is continuous.

Ex ④ Where is $f(x) = \sin^{-1}(x^2)$ continuous?

By Theorem 7 we just need to find the domain of this function.

First, look at $y = \sin x$



The range of $y = \sin x$ is $[-1, 1]$.

This means domain of $\sin^{-1}(x)$ is $[-1, 1]$.

So, for $f(x) = \sin^{-1}(x^2)$ we must have

$$-1 \leq x^2 \leq 1$$

or $-1 \leq x \leq 1$ is the domain of $f(x)$

